



ALLIAN
UNIVERSITY

Private University Estd. in Karnataka State by Act No. 34

Project Report

“Collection of IMDb top 5000 movies data”

By

Piyush Tandale

Department of Computer Application

Alliance University Chandapura — Anekal Main Road,

Anekal Bengaluru — 562 106

April 2025

Project Overview

This project analyzes a dataset of popular movies from IMDb. The data includes movie metadata such as title, release year, rating, votes, genre, runtime, and associated crew members (directors and writers). Through data visualization and statistical exploration, the project seeks to uncover patterns and trends in movie popularity and quality.

Introduction:

The dataset provided appears to be a collection of IMDb movie data, containing information about popular movies. Each row represents a movie and includes the following attributes:

- `tconst`: Unique identifier for the movie (IMDb ID).
- `primaryTitle`: The title of the movie.
- `startYear`: The year the movie was released.
- `rank`: The ranking of the movie based on its popularity or rating.
- `averageRating`: The average IMDb rating of the movie.
- `numVotes`: The number of votes the movie has received on IMDb.
- `runtimeMinutes`: The runtime of the movie in minutes.
- `directors`: The director(s) of the movie.
- `writers`: The writer(s) of the movie.
- `genres`: The genres associated with the movie (e.g., Drama, Action).
- `IMDbLink`: A clickable link to the movie's IMDb page. `Title_IMDb_Link`: Another clickable link to the movie's IMDb page with the title.

Project Goals

1. **Understand Key Attributes** - Gain insights into movie characteristics that correlate with higher IMDb ratings.
2. **Visualize Top Performers** - Identify and visualize the top 10 highest-rated movies.
3. **Explore Genre Trends** - Examine the distribution of movie genres to identify the most common types.
4. **Runtime Analysis** - Investigate the distribution of movie lengths and assess if runtime has any relationship with ratings or genre.
5. **Use Visualizations** - Employ bar plots, pie charts, and histograms for a more intuitive understanding of the data.

Challenges

- **Missing or Incomplete Data:** Some entries may have missing values (especially in runtime or crew information), which could impact analysis accuracy.
- **Genre Complexity:** Movies often belong to multiple genres, but they may be listed as comma-separated strings, making precise categorization difficult.
- **Outliers:** Extremely high or low values in runtime or ratings might skew results and require careful handling.
- **Data Cleaning:** Ensuring the dataset is clean and well-structured for analysis took effort, particularly due to formatting in fields like directors, genres, or IMDb links.

```
import pandas as pd
```

```
df=pd.read_csv(r"C:\Users\piyus\Desktop\ids project\results_with_crew.csv")
```

```
df
```

	tconst	primaryTitle	startYear	\
0	tt0111161	The Shawshank Redemption	1994	
1	tt0068646	The Godfather	1972	
2	tt0468569	The Dark Knight	2008	
3	tt0167260	The Lord of the Rings: The Return of the King	2003	
4	tt0108052	Schindler's List	1993	
...	...	...	...	
4995	tt0880578	Untraceable	2008	
4996	tt27459160	Teri Baaton Mein Aisa Uljha Jiya	2024	
4997	tt3174376	Before I Wake	2016	
4998	tt5177088	The Girl in the Spider's Web	2018	
4999	tt8368408	Gunpowder Milkshake	2021	

	rank	averageRating	numVotes	runtimeMinutes	directors
\					
0	1	9.3	3029407	142	Frank Darabont
1	2	9.2	2114342	175	Francis Ford Coppola
2	3	9.0	3005759	152	Christopher Nolan
3	4	9.0	2068876	201	Peter Jackson
4	5	9.0	1515877	195	Steven Spielberg
...	...	...	...	...	...
4995	4996	6.2	54012	101	Gregory Hoblit
4996	4997	6.2	53979	141	Amit Joshi, Aradhana Sah
4997	4998	6.2	53934	97	Mike Flanagan
4998	4999	6.1	53870	115	Fede Alvarez

4999 5000 6.1 53710 114 Navot Papushado

writers \

0 Stephen King, Frank Darabont
1 Mario Puzo, Francis Ford Coppola
2 Jonathan Nolan, Christopher Nolan, David S. Go...
3 J.R.R. Tolkien, Fran Walsh, Philippa Boyens, P...
4 Thomas Keneally, Steven Zaillian
...
4995 Robert Fyvolent, Mark Brinker, Allison Burnett
4996 Amit Joshi, Aradhana Sah
4997 Mike Flanagan, Jeff Howard
4998 Jay Basu, Fede Alvarez, Steven Knight, Stieg L...
4999 Navot Papushado, Ehud Lavski

genres \

0 Drama
1 Crime, Drama
2 Action, Crime, Drama
3 Adventure, Drama, Fantasy
4 Biography, Drama, History
...
4995 Crime, Mystery, Thriller
4996 Comedy, Drama, Romance
4997 Drama, Fantasy, Horror
4998 Action, Adventure, Crime
4999 Action, Crime, Thriller

IMDbLink \

0 <a href="https://www.imdb.com/title/tt0111161"...
1 <a href="https://www.imdb.com/title/tt0068646"...
2 <a href="https://www.imdb.com/title/tt0468569"...
3 <a href="https://www.imdb.com/title/tt0167260"...
4 <a href="https://www.imdb.com/title/tt0108052"...
...
4995 <a href="https://www.imdb.com/title/tt0880578"...
4996 <a href="https://www.imdb.com/title/tt27459160...
4997 <a href="https://www.imdb.com/title/tt3174376"...
4998 <a href="https://www.imdb.com/title/tt5177088"...
4999 <a href="https://www.imdb.com/title/tt8368408"...

Title_IMDb_Link

0 <a href="https://www.imdb.com/title/tt0111161"...
1 <a href="https://www.imdb.com/title/tt0068646"...
2 <a href="https://www.imdb.com/title/tt0468569"...
3 <a href="https://www.imdb.com/title/tt0167260"...
4 <a href="https://www.imdb.com/title/tt0108052"...
...
4995 <a href="https://www.imdb.com/title/tt0880578"...

```

4996 <a href="https://www.imdb.com/title/tt27459160..."
4997 <a href="https://www.imdb.com/title/tt3174376"..."
4998 <a href="https://www.imdb.com/title/tt5177088"..."
4999 <a href="https://www.imdb.com/title/tt8368408"..."

```

[5000 rows x 12 columns]

```
df.head()
```

	tconst	primaryTitle	startYear	rank
\				
0	tt0111161	The Shawshank Redemption	1994	1
1	tt0068646	The Godfather	1972	2
2	tt0468569	The Dark Knight	2008	3
3	tt0167260	The Lord of the Rings: The Return of the King	2003	4
4	tt0108052	Schindler's List	1993	5

	averageRating	numVotes	runtimeMinutes	directors	\
0	9.3	3029407	142	Frank Darabont	
1	9.2	2114342	175	Francis Ford Coppola	
2	9.0	3005759	152	Christopher Nolan	
3	9.0	2068876	201	Peter Jackson	
4	9.0	1515877	195	Steven Spielberg	

	writers	\
0	Stephen King, Frank Darabont	
1	Mario Puzo, Francis Ford Coppola	
2	Jonathan Nolan, Christopher Nolan, David S. Go...	
3	J.R.R. Tolkien, Fran Walsh, Philippa Boyens, P...	
4	Thomas Keneally, Steven Zaillian	

	genres	\
0	Drama	
1	Crime, Drama	
2	Action, Crime, Drama	
3	Adventure, Drama, Fantasy	
4	Biography, Drama, History	

	IMDbLink	\
0	<a href="https://www.imdb.com/title/tt0111161"...	
1	<a href="https://www.imdb.com/title/tt0068646"...	
2	<a href="https://www.imdb.com/title/tt0468569"...	
3	<a href="https://www.imdb.com/title/tt0167260"...	
4	<a href="https://www.imdb.com/title/tt0108052"...	

	Title_IMDb_Link
0	<a href="https://www.imdb.com/title/tt0111161"...
1	<a href="https://www.imdb.com/title/tt0068646"...
2	<a href="https://www.imdb.com/title/tt0468569"...

```
3 <a href="https://www.imdb.com/title/tt0167260"...
4 <a href="https://www.imdb.com/title/tt0108052"...
```

```
df.tail()
```

	tconst	primaryTitle	startYear	rank	\
4995	tt0880578	Untraceable	2008	4996	
4996	tt27459160	Teri Baaton Mein Aisa Uljha Jiya	2024	4997	
4997	tt3174376	Before I Wake	2016	4998	
4998	tt5177088	The Girl in the Spider's Web	2018	4999	
4999	tt8368408	Gunpowder Milkshake	2021	5000	

	averageRating	numVotes	runtimeMinutes	directors	\
4995	6.2	54012	101	Gregory Hoblit	
4996	6.2	53979	141	Amit Joshi, Aradhana Sah	
4997	6.2	53934	97	Mike Flanagan	
4998	6.1	53870	115	Fede Alvarez	
4999	6.1	53710	114	Navot Papushado	

	writers	\
4995	Robert Fyvolent, Mark Brinker, Allison Burnett	
4996	Amit Joshi, Aradhana Sah	
4997	Mike Flanagan, Jeff Howard	
4998	Jay Basu, Fede Alvarez, Steven Knight, Stieg L...	
4999	Navot Papushado, Ehud Lavski	

	genres	\
4995	Crime, Mystery, Thriller	
4996	Comedy, Drama, Romance	
4997	Drama, Fantasy, Horror	
4998	Action, Adventure, Crime	
4999	Action, Crime, Thriller	

	IMDbLink	\
4995	<a href="https://www.imdb.com/title/tt0880578"...	
4996	<a href="https://www.imdb.com/title/tt27459160..."	
4997	<a href="https://www.imdb.com/title/tt3174376"...	
4998	<a href="https://www.imdb.com/title/tt5177088"...	
4999	<a href="https://www.imdb.com/title/tt8368408"...	

	Title_IMDb_Link
4995	<a href="https://www.imdb.com/title/tt0880578"...
4996	<a href="https://www.imdb.com/title/tt27459160..."
4997	<a href="https://www.imdb.com/title/tt3174376"...
4998	<a href="https://www.imdb.com/title/tt5177088"...
4999	<a href="https://www.imdb.com/title/tt8368408"...

```
df.info
```

```

<bound method DataFrame.info of          tconst
primaryTitle startYear \
0      tt0111161          The Shawshank Redemption      1994
1      tt0068646          The Godfather                1972
2      tt0468569          The Dark Knight              2008
3      tt0167260  The Lord of the Rings: The Return of the King  2003
4      tt0108052          Schindler's List              1993
...      ...
4995    tt0880578          Untraceable                  2008
4996    tt27459160      Teri Baaton Mein Aisa Uljha Jiya      2024
4997    tt3174376          Before I Wake                2016
4998    tt5177088      The Girl in the Spider's Web          2018
4999    tt8368408          Gunpowder Milkshake            2021

```

```

rank averageRating numVotes runtimeMinutes directors
\
0      1          9.3   3029407          142      Frank Darabont
1      2          9.2   2114342          175      Francis Ford Coppola
2      3          9.0   3005759          152      Christopher Nolan
3      4          9.0   2068876          201      Peter Jackson
4      5          9.0   1515877          195      Steven Spielberg
...      ...
4995  4996          6.2    54012          101      Gregory Hoblit
4996  4997          6.2    53979          141  Amit Joshi, Aradhana Sah
4997  4998          6.2    53934           97      Mike Flanagan
4998  4999          6.1    53870          115      Fede Alvarez
4999  5000          6.1    53710          114      Navot Papushado

```

```

writers \
0      Stephen King, Frank Darabont
1      Mario Puzo, Francis Ford Coppola
2      Jonathan Nolan, Christopher Nolan, David S. Go...
3      J.R.R. Tolkien, Fran Walsh, Philippa Boyens, P...
4      Thomas Keneally, Steven Zaillian
...
4995    Robert Fyvolent, Mark Brinker, Allison Burnett
4996          Amit Joshi, Aradhana Sah
4997          Mike Flanagan, Jeff Howard
4998  Jay Basu, Fede Alvarez, Steven Knight, Stieg L...
4999          Navot Papushado, Ehud Lavski

```

```

genres \
0      Drama
1      Crime, Drama
2      Action, Crime, Drama
3      Adventure, Drama, Fantasy
4      Biography, Drama, History
...
4995    Crime, Mystery, Thriller
4996    Comedy, Drama, Romance

```

```

4997    Drama, Fantasy, Horror
4998    Action, Adventure, Crime
4999    Action, Crime, Thriller

```

```

                                IMDbLink \
0      <a href="https://www.imdb.com/title/tt0111161"...
1      <a href="https://www.imdb.com/title/tt0068646"...
2      <a href="https://www.imdb.com/title/tt0468569"...
3      <a href="https://www.imdb.com/title/tt0167260"...
4      <a href="https://www.imdb.com/title/tt0108052"...
...
4995   <a href="https://www.imdb.com/title/tt0880578"...
4996   <a href="https://www.imdb.com/title/tt27459160...
4997   <a href="https://www.imdb.com/title/tt3174376"...
4998   <a href="https://www.imdb.com/title/tt5177088"...
4999   <a href="https://www.imdb.com/title/tt8368408"...

```

```

                                Title_IMDb_Link
0      <a href="https://www.imdb.com/title/tt0111161"...
1      <a href="https://www.imdb.com/title/tt0068646"...
2      <a href="https://www.imdb.com/title/tt0468569"...
3      <a href="https://www.imdb.com/title/tt0167260"...
4      <a href="https://www.imdb.com/title/tt0108052"...
...
4995   <a href="https://www.imdb.com/title/tt0880578"...
4996   <a href="https://www.imdb.com/title/tt27459160...
4997   <a href="https://www.imdb.com/title/tt3174376"...
4998   <a href="https://www.imdb.com/title/tt5177088"...
4999   <a href="https://www.imdb.com/title/tt8368408"...

```

```
[5000 rows x 12 columns]>
```

```
df.describe()
```

	startYear	rank	averageRating	numVotes	runtimeMinutes
count	5000.000000	5000.000000	5000.000000	5.000000e+03	5000.000000
mean	2002.021200	2500.500000	7.137140	1.630208e+05	114.56380
std	18.262844	1443.520003	0.597229	2.403863e+05	23.21629
min	1915.000000	1.000000	5.900000	2.500000e+04	25.00000
25%	1994.000000	1250.750000	6.700000	4.008425e+04	99.00000
50%	2007.000000	2500.500000	7.100000	7.438750e+04	111.00000
75%	2015.000000	3750.250000	7.600000	1.780925e+05	126.00000
max	2025.000000	5000.000000	9.300000	3.029407e+06	374.00000

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

```

```
# Load the dataset
```

```
df = pd.read_csv(r"C:\Users\piyus\Desktop\ids project\results_with_crew.csv")
```


df

	tconst	primaryTitle	startYear	\
0	tt01111161	The Shawshank Redemption	1994	
1	tt0068646	The Godfather	1972	
2	tt0468569	The Dark Knight	2008	
3	tt0167260	The Lord of the Rings: The Return of the King	2003	
4	tt0108052	Schindler's List	1993	
...	...	...	...	
4995	tt0880578	Untraceable	2008	
4996	tt27459160	Teri Baaton Mein Aisa Uljha Jiya	2024	
4997	tt3174376	Before I Wake	2016	
4998	tt5177088	The Girl in the Spider's Web	2018	
4999	tt8368408	Gunpowder Milkshake	2021	

	rank	averageRating	numVotes	runtimeMinutes	directors
\					
0	1	9.3	3029407	142	Frank Darabont
1	2	9.2	2114342	175	Francis Ford Coppola
2	3	9.0	3005759	152	Christopher Nolan
3	4	9.0	2068876	201	Peter Jackson
4	5	9.0	1515877	195	Steven Spielberg
...	...	...	...	...	...
4995	4996	6.2	54012	101	Gregory Hoblit
4996	4997	6.2	53979	141	Amit Joshi, Aradhana Sah
4997	4998	6.2	53934	97	Mike Flanagan
4998	4999	6.1	53870	115	Fede Alvarez
4999	5000	6.1	53710	114	Navot Papushado

	writers	\
0	Stephen King, Frank Darabont	
1	Mario Puzo, Francis Ford Coppola	
2	Jonathan Nolan, Christopher Nolan, David S. Go...	
3	J.R.R. Tolkien, Fran Walsh, Philippa Boyens, P...	
4	Thomas Keneally, Steven Zaillian	
...	...	
4995	Robert Fyvolent, Mark Brinker, Allison Burnett	
4996	Amit Joshi, Aradhana Sah	
4997	Mike Flanagan, Jeff Howard	
4998	Jay Basu, Fede Alvarez, Steven Knight, Stieg L...	
4999	Navot Papushado, Ehud Lavski	

	genres	\
0	Drama	
1	Crime, Drama	
2	Action, Crime, Drama	
3	Adventure, Drama, Fantasy	
4	Biography, Drama, History	
...	...	
4995	Crime, Mystery, Thriller	

```

4996      Comedy, Drama, Romance
4997      Drama, Fantasy, Horror
4998      Action, Adventure, Crime
4999      Action, Crime, Thriller

```

```

                                IMDbLink \
0      <a href="https://www.imdb.com/title/tt0111161"...
1      <a href="https://www.imdb.com/title/tt0068646"...
2      <a href="https://www.imdb.com/title/tt0468569"...
3      <a href="https://www.imdb.com/title/tt0167260"...
4      <a href="https://www.imdb.com/title/tt0108052"...
...
4995   <a href="https://www.imdb.com/title/tt0880578"...
4996   <a href="https://www.imdb.com/title/tt27459160...
4997   <a href="https://www.imdb.com/title/tt3174376"...
4998   <a href="https://www.imdb.com/title/tt5177088"...
4999   <a href="https://www.imdb.com/title/tt8368408"...

```

```

                                Title_IMDb_Link
0      <a href="https://www.imdb.com/title/tt0111161"...
1      <a href="https://www.imdb.com/title/tt0068646"...
2      <a href="https://www.imdb.com/title/tt0468569"...
3      <a href="https://www.imdb.com/title/tt0167260"...
4      <a href="https://www.imdb.com/title/tt0108052"...
...
4995   <a href="https://www.imdb.com/title/tt0880578"...
4996   <a href="https://www.imdb.com/title/tt27459160...
4997   <a href="https://www.imdb.com/title/tt3174376"...
4998   <a href="https://www.imdb.com/title/tt5177088"...
4999   <a href="https://www.imdb.com/title/tt8368408"...

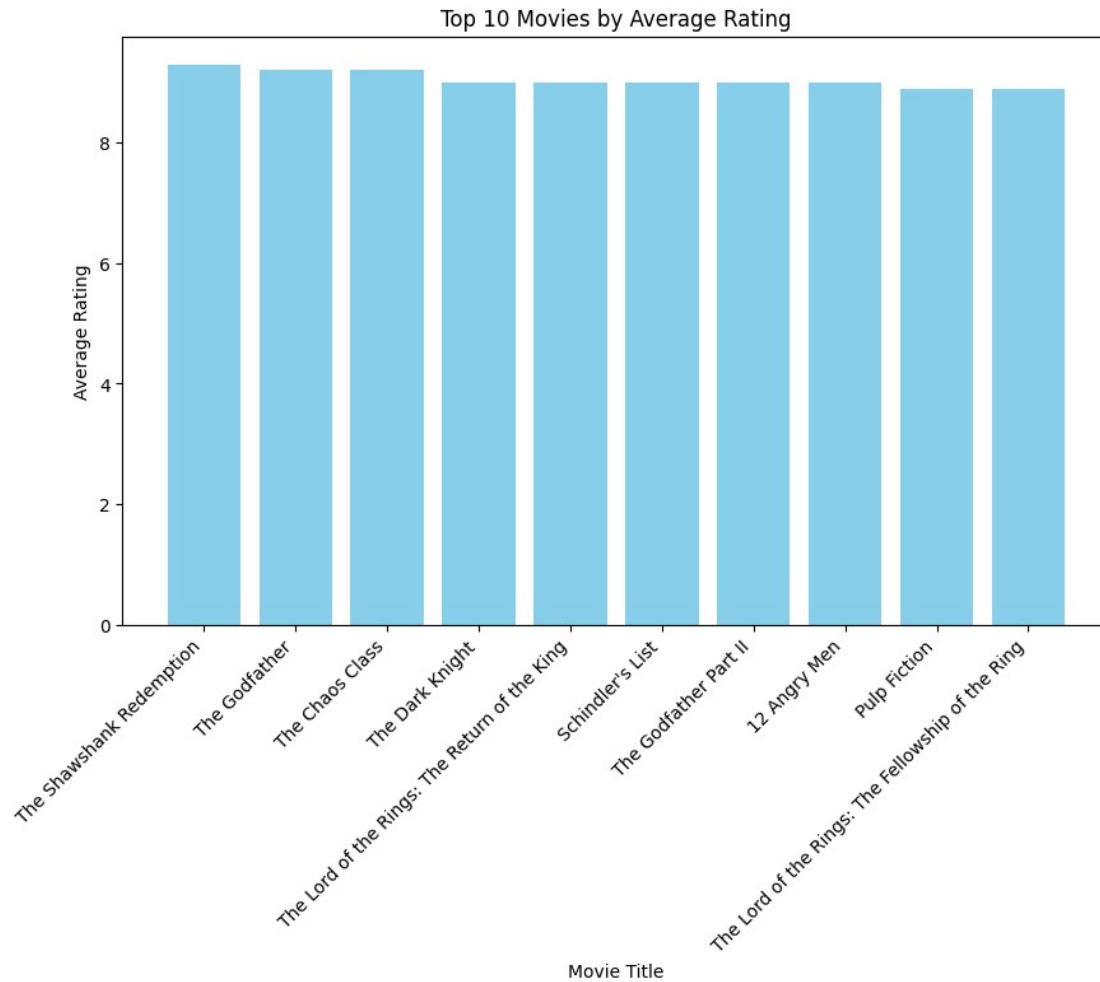
```

```
[5000 rows x 12 columns]
```

```

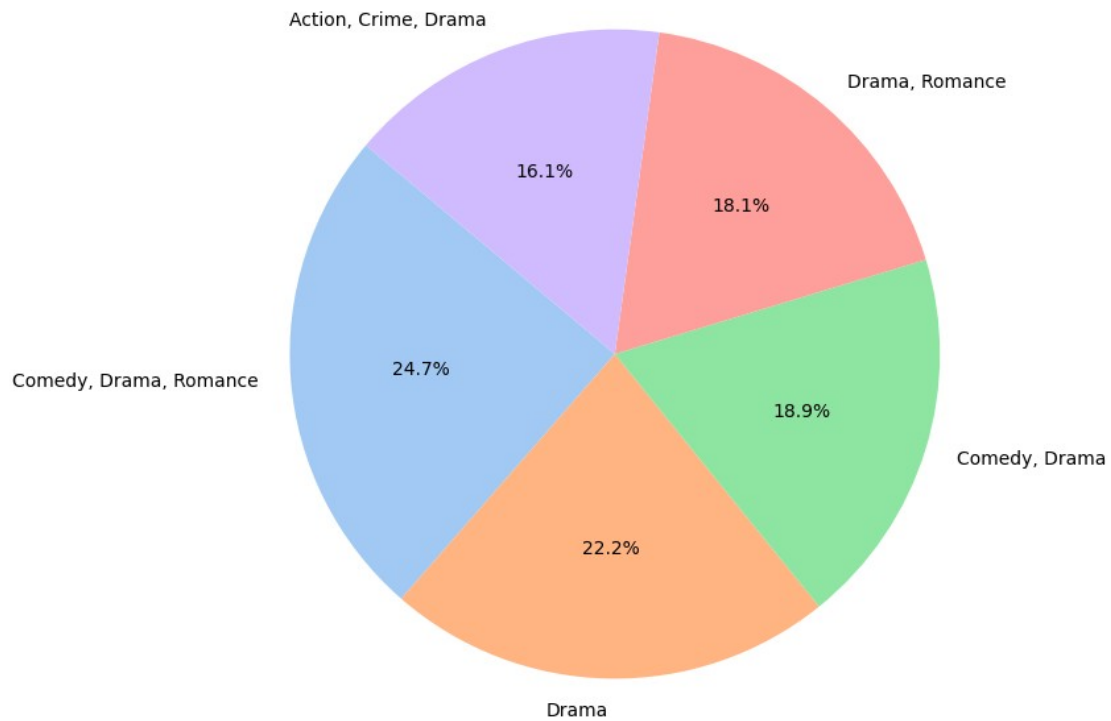
# Bar Plot: Top 10 movies by average rating
top_10_movies = df.nlargest(10, 'averageRating')
plt.figure(figsize=(10, 6))
plt.bar(top_10_movies['primaryTitle'], top_10_movies['averageRating'],
color='skyblue')
plt.xticks(rotation=45, ha='right')
plt.title('Top 10 Movies by Average Rating')
plt.xlabel('Movie Title')
plt.ylabel('Average Rating')
plt.show()

```

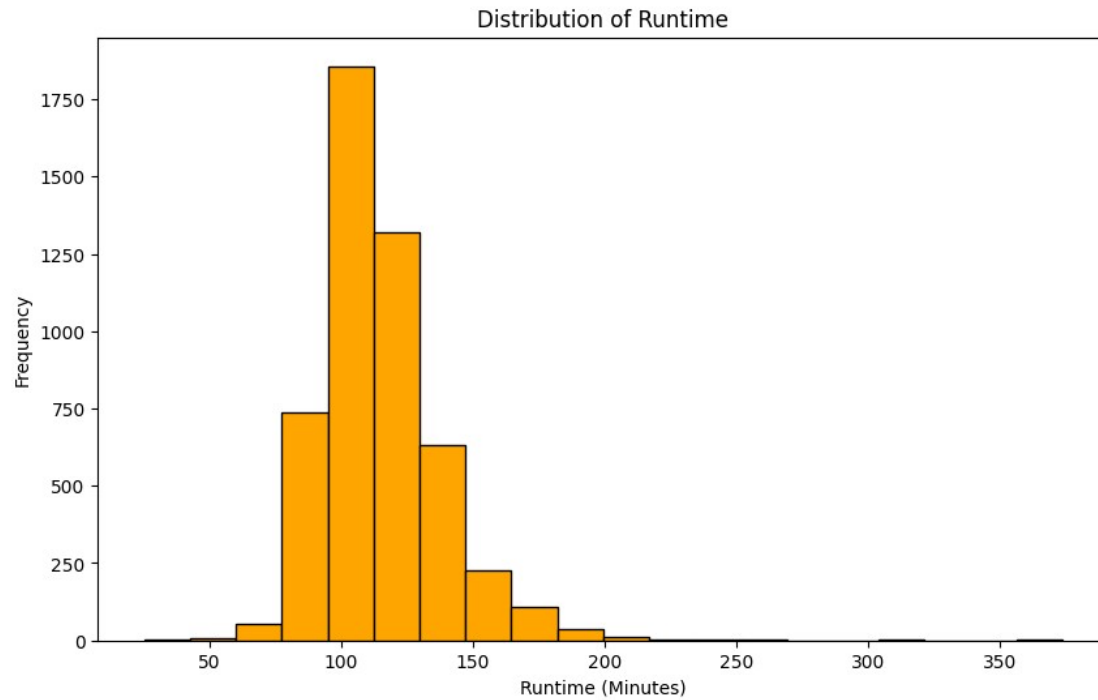


```
# Pie Chart: Distribution of genres
genre_counts = df['genres'].value_counts().head(5) # Top 5 genres
plt.figure(figsize=(8, 8))
plt.pie(genre_counts, labels=genre_counts.index, autopct='%1.1f%%',
startangle=140, colors=sns.color_palette("pastel"))
plt.title('Top 5 Genres Distribution')
plt.show()
```

Top 5 Genres Distribution



```
# Histogram: Distribution of runtime
plt.figure(figsize=(10, 6))
plt.hist(df['runtimeMinutes'].dropna(), bins=20, color='orange',
edgecolor='black')
plt.title('Distribution of Runtime')
plt.xlabel('Runtime (Minutes)')
plt.ylabel('Frequency')
plt.show()
```



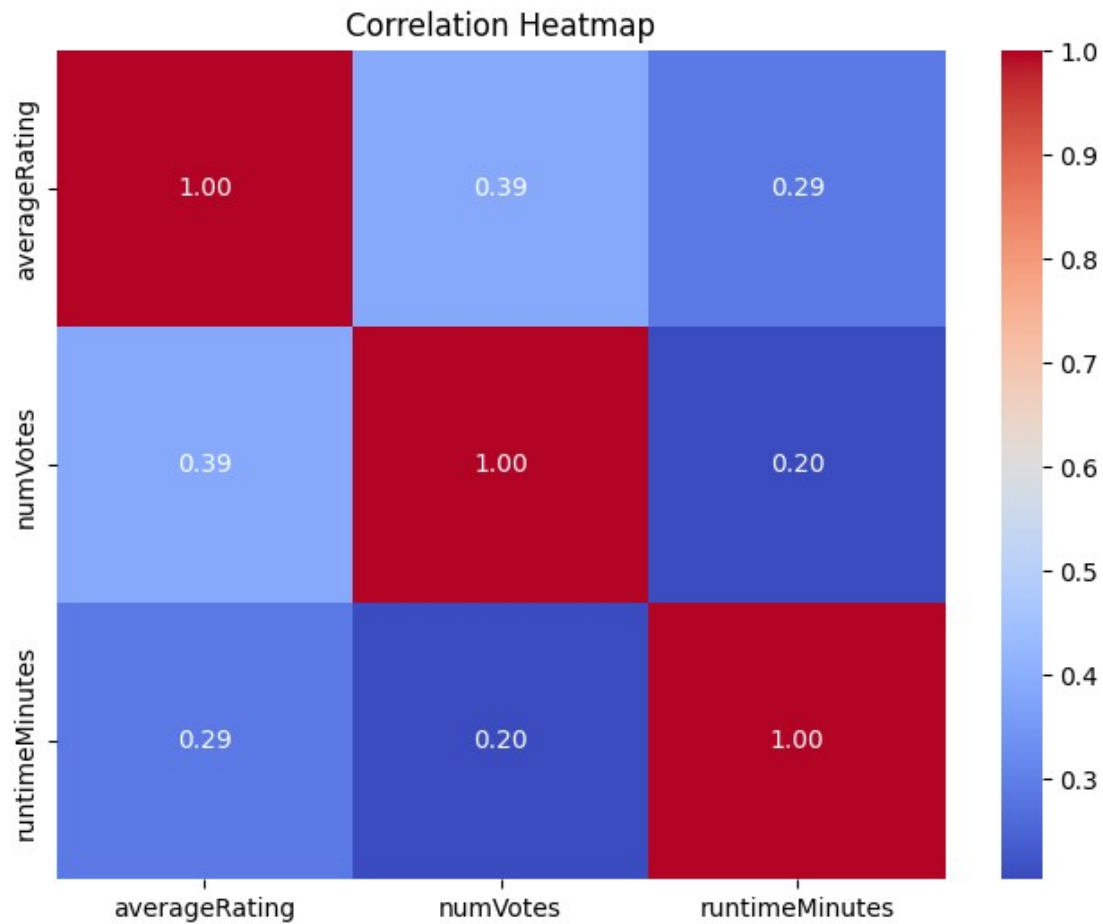
```
# Correlation Matrix
```

```
correlation_matrix = df[['averageRating', 'numVotes',  
    'runtimeMinutes']].corr()  
print(correlation_matrix)
```

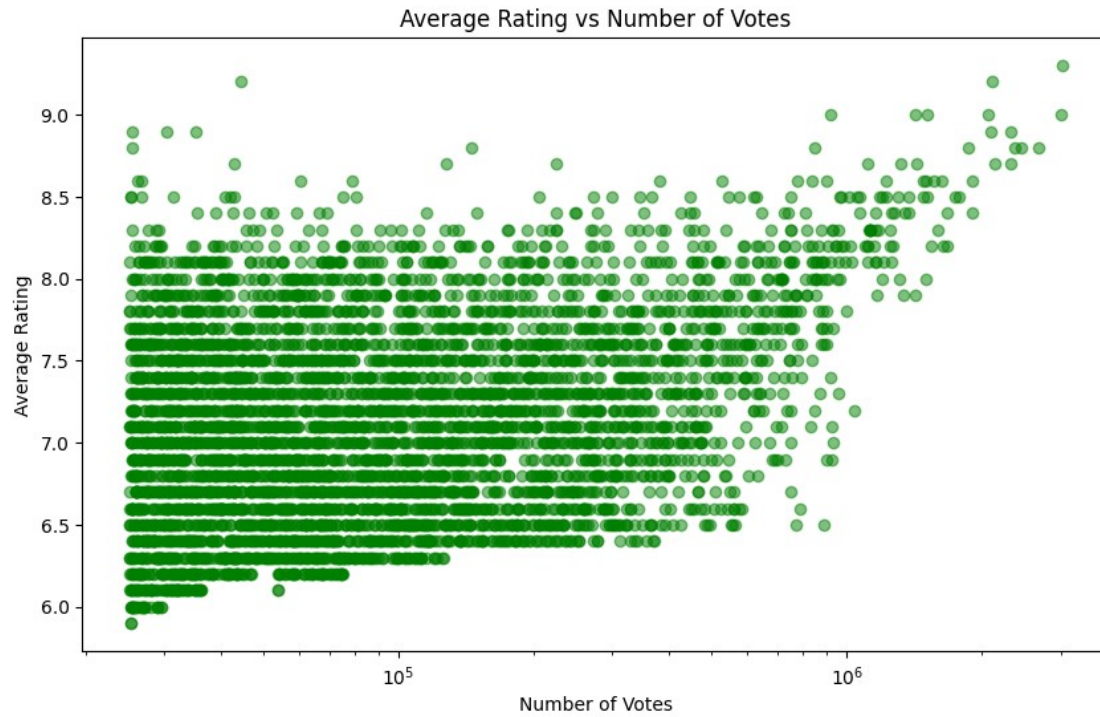
	averageRating	numVotes	runtimeMinutes
averageRating	1.000000	0.387595	0.287495
numVotes	0.387595	1.000000	0.203039
runtimeMinutes	0.287495	0.203039	1.000000

```
# Correlation Heatmap
```

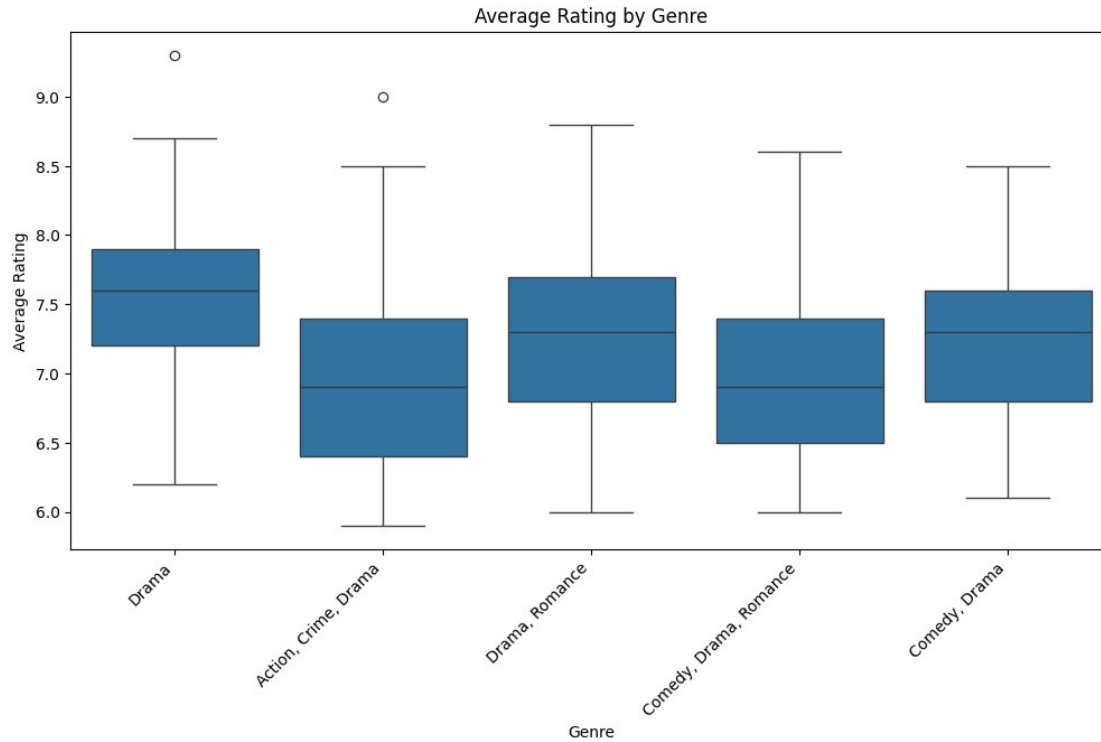
```
plt.figure(figsize=(8, 6))  
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f')  
plt.title('Correlation Heatmap')  
plt.show()
```



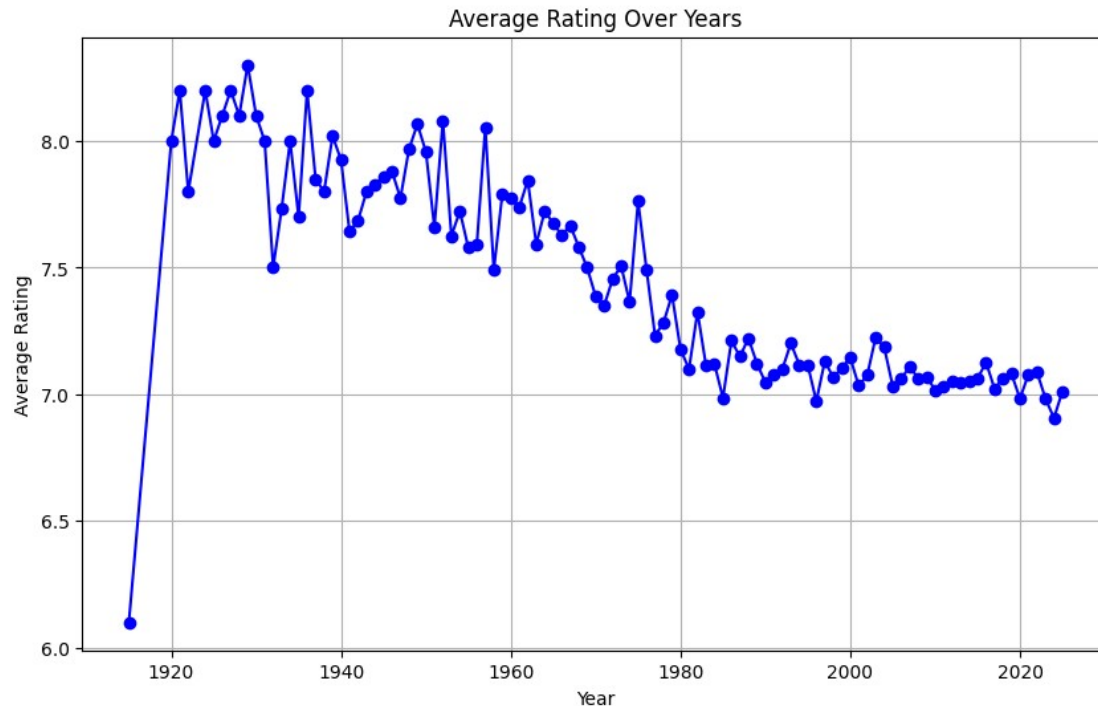
```
# Scatter Plot: Average Rating vs Number of Votes
plt.figure(figsize=(10, 6))
plt.scatter(df['numVotes'], df['averageRating'], alpha=0.5, color='green')
plt.title('Average Rating vs Number of Votes')
plt.xlabel('Number of Votes')
plt.ylabel('Average Rating')
plt.xscale('log') # Log scale for better visualization
plt.show()
```



```
# Boxplot: Average Rating by Genre
plt.figure(figsize=(12, 6))
sns.boxplot(x='genres', y='averageRating',
data=df[df['genres'].isin(genre_counts.index)])
plt.xticks(rotation=45, ha='right')
plt.title('Average Rating by Genre')
plt.xlabel('Genre')
plt.ylabel('Average Rating')
plt.show()
```



```
# Time-Series Analysis: Average Rating over Years
df['startYear'] = pd.to_numeric(df['startYear'], errors='coerce') # Ensure
numeric year
yearly_avg_rating = df.groupby('startYear')['averageRating'].mean().dropna()
plt.figure(figsize=(10, 6))
plt.plot(yearly_avg_rating.index, yearly_avg_rating.values, marker='o',
linestyle='--', color='blue')
plt.title('Average Rating Over Years')
plt.xlabel('Year')
plt.ylabel('Average Rating')
plt.grid()
plt.show()
```

```
# Data Cleaning & Preprocessing
# Handle missing values and ensure correct data types
df['startYear'] = pd.to_numeric(df['startYear'], errors='coerce') # Convert
startYear to numeric
df['runtimeMinutes'] = pd.to_numeric(df['runtimeMinutes'], errors='coerce')
# Convert runtimeMinutes to numeric
df['averageRating'] = pd.to_numeric(df['averageRating'], errors='coerce') #
Convert averageRating to numeric
df['numVotes'] = pd.to_numeric(df['numVotes'], errors='coerce') # Convert
numVotes to numeric
```

```
# Drop rows with missing critical values
```

```
df = df.dropna(subset=['averageRating', 'numVotes', 'runtimeMinutes',
'startYear'])
```

```
df
```

	tconst	primaryTitle	startYear	\
0	tt0111161	The Shawshank Redemption	1994	
1	tt0068646	The Godfather	1972	
2	tt0468569	The Dark Knight	2008	
3	tt0167260	The Lord of the Rings: The Return of the King	2003	
4	tt0108052	Schindler's List	1993	
...	...	...	...	
4995	tt0880578	Untraceable	2008	
4996	tt27459160	Teri Baaton Mein Aisa Uljha Jiya	2024	
4997	tt3174376	Before I Wake	2016	
4998	tt5177088	The Girl in the Spider's Web	2018	
4999	tt8368408	Gunpowder Milkshake	2021	

	rank	averageRating	numVotes	runtimeMinutes	directors
\					
0	1	9.3	3029407	142	Frank Darabont
1	2	9.2	2114342	175	Francis Ford Coppola
2	3	9.0	3005759	152	Christopher Nolan
3	4	9.0	2068876	201	Peter Jackson
4	5	9.0	1515877	195	Steven Spielberg
...	...	...	...	...	...
4995	4996	6.2	54012	101	Gregory Hoblit
4996	4997	6.2	53979	141	Amit Joshi, Aradhana Sah
4997	4998	6.2	53934	97	Mike Flanagan
4998	4999	6.1	53870	115	Fede Alvarez
4999	5000	6.1	53710	114	Navot Papushado

	writers
\	
0	Stephen King, Frank Darabont
1	Mario Puzo, Francis Ford Coppola
2	Jonathan Nolan, Christopher Nolan, David S. Go...
3	J.R.R. Tolkien, Fran Walsh, Philippa Boyens, P...
4	Thomas Keneally, Steven Zaillian
...	...
4995	Robert Fyvolent, Mark Brinker, Allison Burnett
4996	Amit Joshi, Aradhana Sah
4997	Mike Flanagan, Jeff Howard
4998	Jay Basu, Fede Alvarez, Steven Knight, Stieg L...
4999	Navot Papushado, Ehud Lavski

	genres
\	
0	Drama
1	Crime, Drama
2	Action, Crime, Drama
3	Adventure, Drama, Fantasy
4	Biography, Drama, History
...	...
4995	Crime, Mystery, Thriller
4996	Comedy, Drama, Romance
4997	Drama, Fantasy, Horror
4998	Action, Adventure, Crime
4999	Action, Crime, Thriller

	IMDbLink
\	
0	<a href="https://www.imdb.com/title/tt0111161"...
1	<a href="https://www.imdb.com/title/tt0068646"...
2	<a href="https://www.imdb.com/title/tt0468569"...
3	<a href="https://www.imdb.com/title/tt0167260"...
4	<a href="https://www.imdb.com/title/tt0108052"...
...	...
4995	<a href="https://www.imdb.com/title/tt0880578"...

```

4996 <a href="https://www.imdb.com/title/tt27459160...
4997 <a href="https://www.imdb.com/title/tt3174376"...
4998 <a href="https://www.imdb.com/title/tt5177088"...
4999 <a href="https://www.imdb.com/title/tt8368408"...

```

```

                                Title_IMDb_Link
0      <a href="https://www.imdb.com/title/tt0111161"...
1      <a href="https://www.imdb.com/title/tt0068646"...
2      <a href="https://www.imdb.com/title/tt0468569"...
3      <a href="https://www.imdb.com/title/tt0167260"...
4      <a href="https://www.imdb.com/title/tt0108052"...
...
4995 <a href="https://www.imdb.com/title/tt0880578"...
4996 <a href="https://www.imdb.com/title/tt27459160...
4997 <a href="https://www.imdb.com/title/tt3174376"...
4998 <a href="https://www.imdb.com/title/tt5177088"...
4999 <a href="https://www.imdb.com/title/tt8368408"...

```

```
[5000 rows x 12 columns]
```

```
# Extract primary genre for simplicity
```

```
df['primaryGenre'] = df['genres'].str.split(',').str[0]
```

```
# EDA with plots
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Distribution of average ratings
```

```
plt.figure(figsize=(10, 6))
```

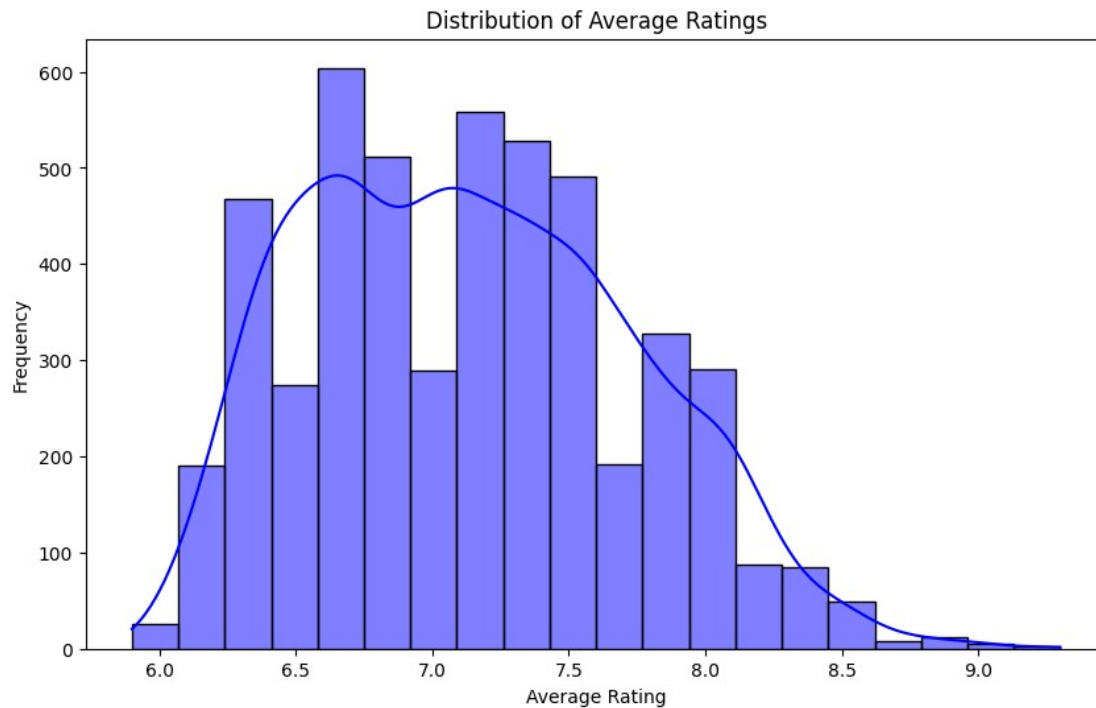
```
sns.histplot(df['averageRating'], bins=20, kde=True, color='blue')
```

```
plt.title('Distribution of Average Ratings')
```

```
plt.xlabel('Average Rating')
```

```
plt.ylabel('Frequency')
```

```
plt.show()
```



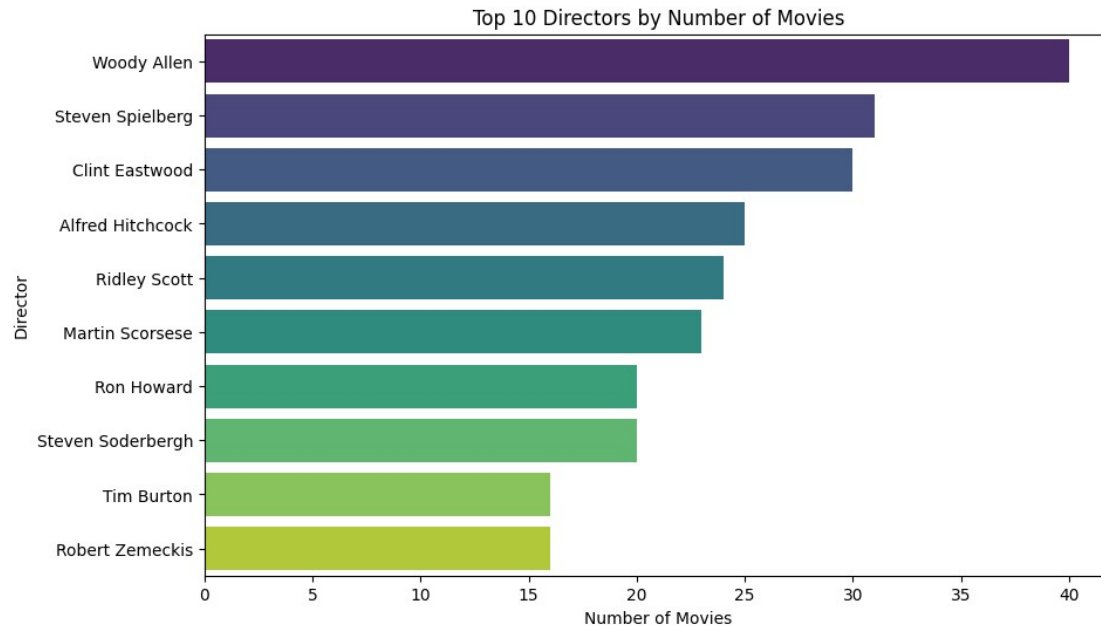
Top 10 directors by number of movies

```
top_directors = df['directors'].value_counts().head(10)
plt.figure(figsize=(10, 6))
sns.barplot(x=top_directors.values, y=top_directors.index, palette='viridis')
plt.title('Top 10 Directors by Number of Movies')
plt.xlabel('Number of Movies')
plt.ylabel('Director')
plt.show()
```

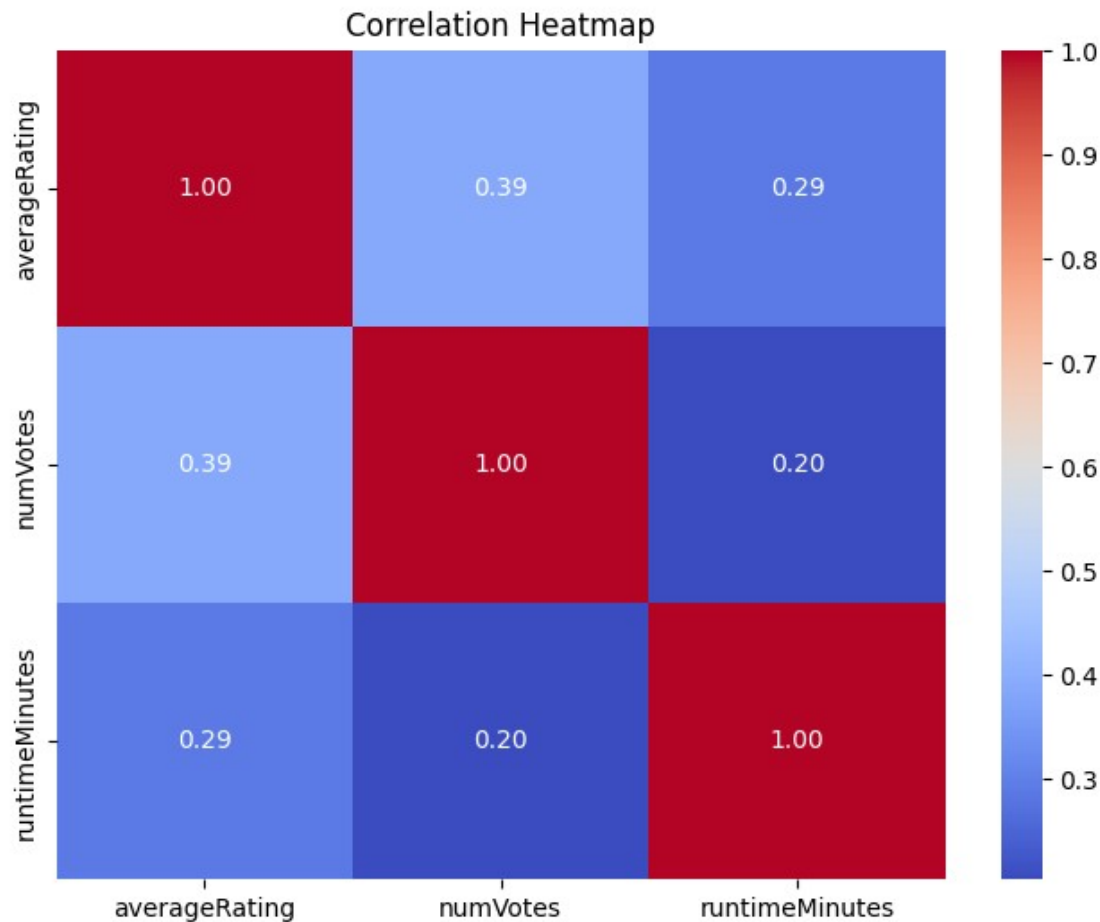
C:\Users\piyus\AppData\Local\Temp\ipykernel_26520\4062912145.py:4:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=top_directors.values, y=top_directors.index,
palette='viridis')
```



```
# Correlation heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(df[['averageRating', 'numVotes', 'runtimeMinutes']].corr(),
            annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```



```
# Feature Engineering
# Create a new feature: Log-transformed numVotes for better scaling
import numpy as np
df['log_numVotes'] = np.log1p(df['numVotes'])

# One-hot encode primaryGenre
df = pd.get_dummies(df, columns=['primaryGenre'], drop_first=True)

# Select features and target
X = df[['runtimeMinutes', 'log_numVotes']] + [col for col in df.columns if
col.startswith('primaryGenre_')]
y = df['averageRating']

# Train-Test Split
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Train a Linear Regression Model
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
```

```
LinearRegression()

# Evaluate the model
from sklearn.metrics import mean_squared_error, r2_score

# Predictions
y_pred = model.predict(X_test)

# Metrics
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error: {mse}")
print(f"R-squared: {r2}")

Mean Squared Error: 0.2848424333814623
R-squared: 0.24909020219910427
```

Conclusion

This project successfully used Python's data analysis libraries (Pandas, Matplotlib, Seaborn) to extract insights from an IMDb movie dataset. It highlighted the characteristics of top-rated films, identified dominant genres, and explored runtime distributions. Despite challenges with data quality and formatting, the analysis provided a deeper understanding of what factors may influence movie ratings and popularity, offering valuable insights for further exploration or recommendation system development.